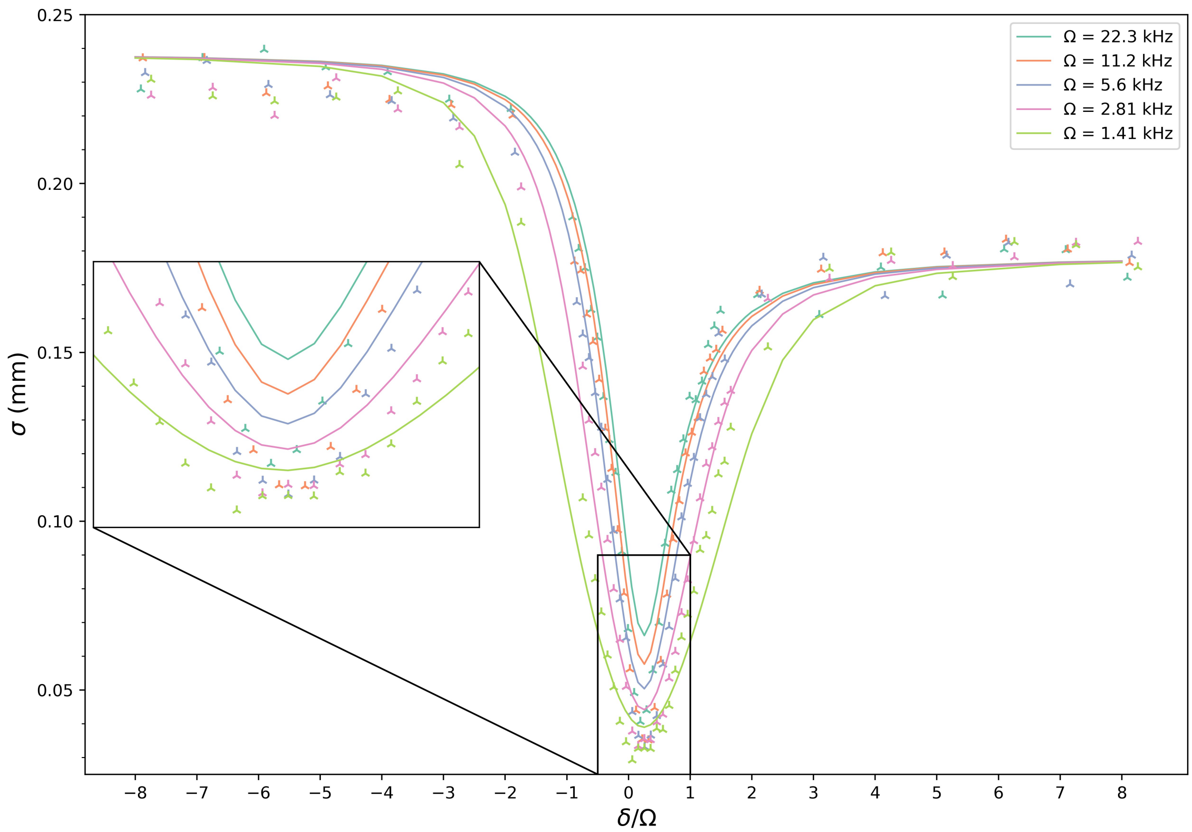
Report GPELab simulations

Simulations with BMF effects

Simulations with BMF effects added in modifying a_{12} (as explained in the "output_analysis.pdf" file) in GPE simulation setting parameters

Entrée [10]:
```
from IPython.display import display, Image

fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GP
display(Image(filename=fileName))
```

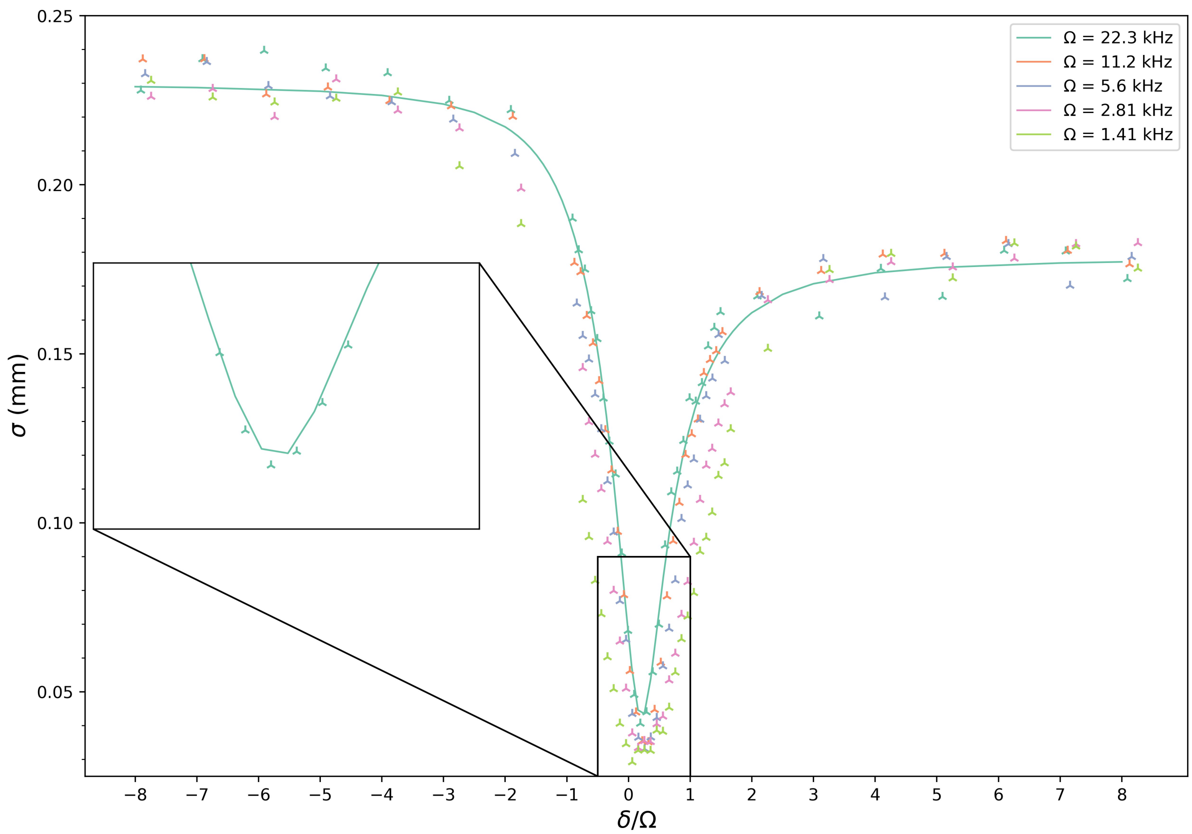


COMMENTS ON THIS LAST PLOT

BMF effects added in the simulation modifying a12
We can appreciate how probably the experimental data do not capture as much BMF effect as we thought. We clearly see that the added BMF correction shifts the simulation results at the minimum of the curve, but now the simulation curves are above the data. In fact we can calculate the min point for all the curves and then evaluate the distance between the two curves with the highest and lowest Rabi frequency. We can in that way compare those distances evaluated for the simulated curves and for the data.
The result is:
SIMULATION $\Delta = 0.027$ mm
EXP DATA $\Delta = 0.0114$ mm

I could try to adjust the scattering length a11 and the density so to fit well the 22kHz curve. And then evaluate all the other curves for those parameters.
In ordet to fit the 22kHz cirve I needed to lower $a_{11} \rightarrow a_{11} - 10a_0$ and to higher the peak density $n_0 = 5.3e9$. The curve is so fitted well (see plot below). The problem is that for those parameters the lower Rabi curves can not be simulated because we so in the attractive region. This is the proof that the data do not have as much BMF effects. In addiction I know that for negative detuning the numper of atoms is lower than at the beginning of the sweep, so I expect the simulaiton to be abowe the data in this region.

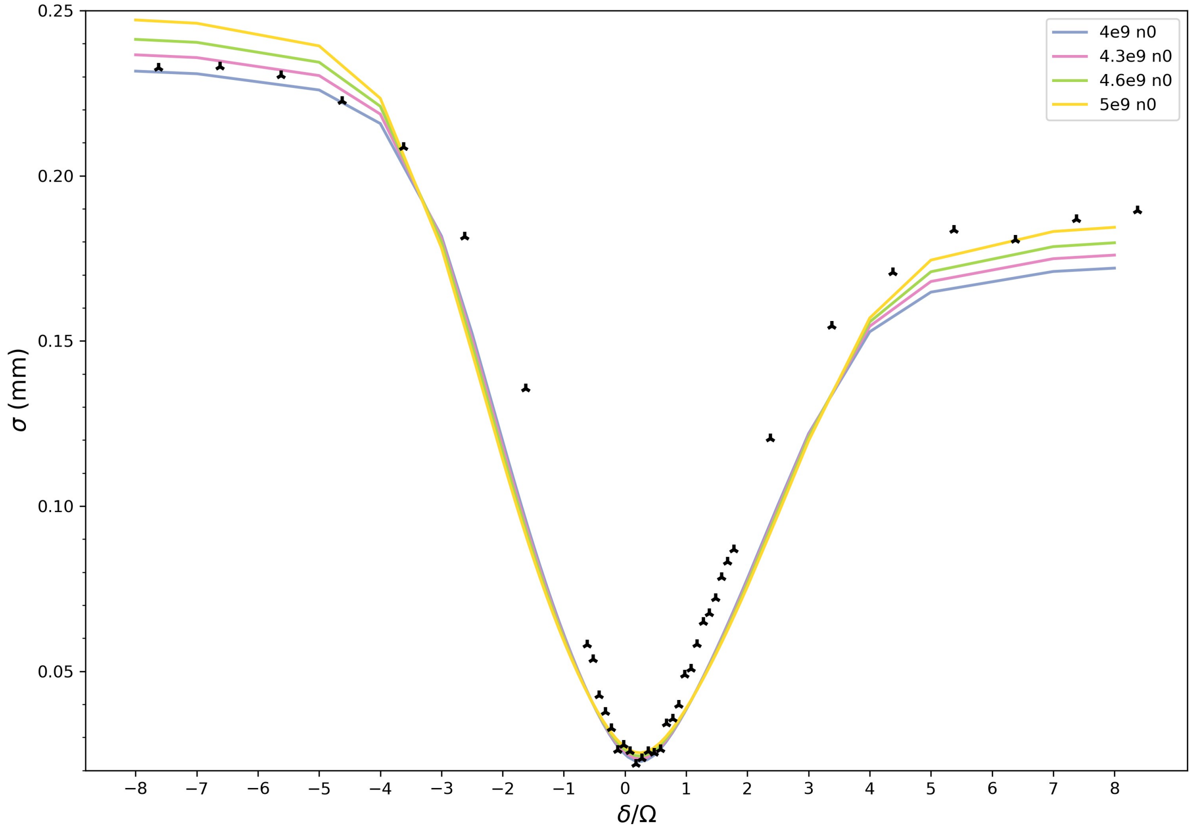
Entrée [13]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GP
display(Image(filename=fileName))
```



MF simulation: Tiemman scatteing lengths

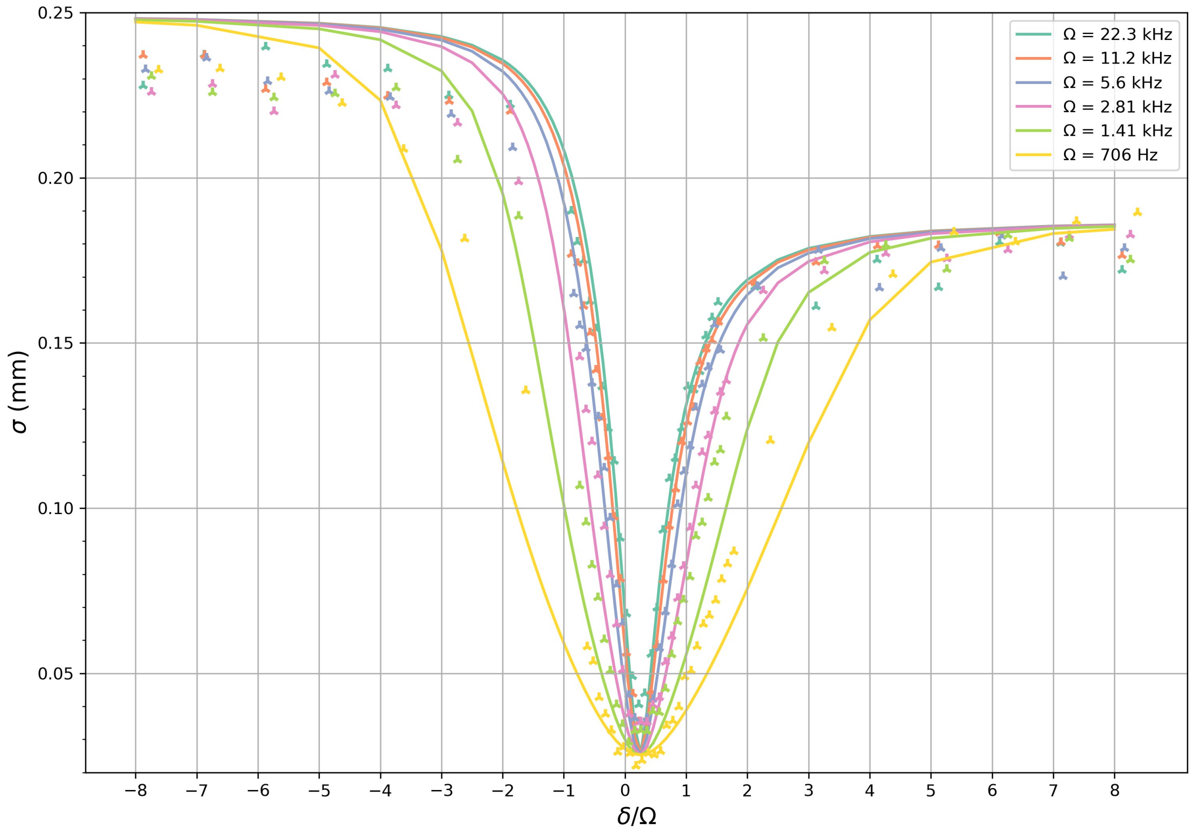
I use Tiemann scatteirng length. I now run different simulations tuning the peak density to fit well the data at big positive detuning (at really low detuning the simulation will be probably higher than the data since we have some losses of atoms in the adiabatic sweep.

Entrée [7]:
```
fileName = r"C:\Users\Sarah\Documents\GitHub\Numerical-simulations\GPELab\plot
display(Image(filename=fileName))
```



Plot the results for each Rabi frequency and for n0=5e9

Entrée [5]:
```
from IPython.display import display, Image
fileName = r"C:\Users\Sarah\Documents\GitHub\Numerical-simulations\GPELab\plot
display(Image(filename=fileName))
```

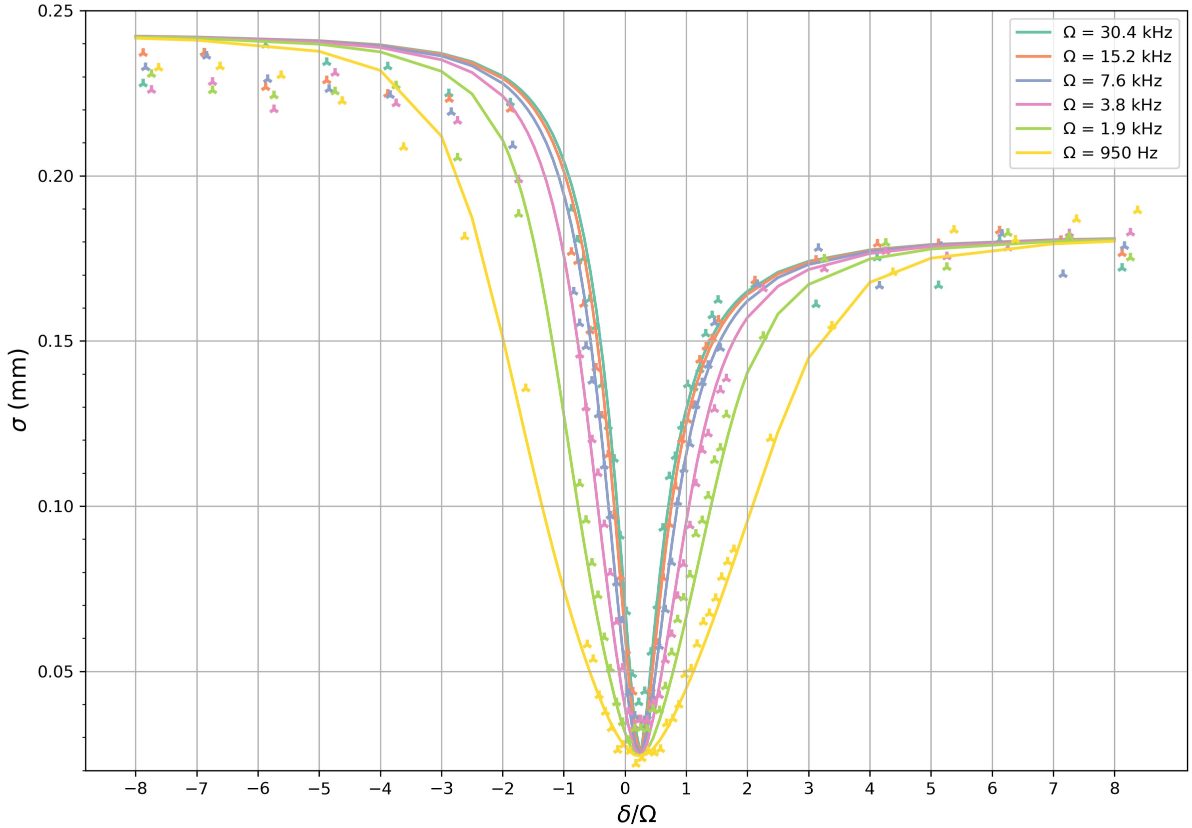


Actually Roy remeasured the Rabi frequencies at the beginning of February 2025 and the Rabi frequency for the 6 curves could be
Omega_values = [30400, 15200, 7600, 3800, 1900, 950] Hz

Let's try to run te sumilation for the 950 Hz curve and for different values of n0.

I select n0 = 4.6e9

Entrée [3]:
```
from IPython.display import display, Image
fileName = r"C:\Users\Sarah\Documents\GitHub\Numerical-simulations\GPELab\plot
display(Image(filename=fileName))
```



Entrée [ ]: